

 @JAIIB CAIIB PREP BY MAHESH SIR



# JAIIB CAIIB PREP



**BFM Module C Concept Unit wise marathon :**

# Treasury Management

**Expert guidance for JAIIB CAIIB examinations**

Follow us on:



Visit our website:

<https://www.jaiibcaiibprep.com>

Download our app:



## Welcome to JAIIB CAIIB Prep

**Your Ultimate Destination  
for Bank Exam Success!**

**What We Offer:**

- ✓ **Concept-Building Lectures**
- ✓ **Daily MCQs & Case Study Practice**
- ✓ **Marathon & Mock Test Series**
- ✓ **Complete Module-Wise Coverage**
- ✓ **Tips, Tricks & Strategy Sessions**



## JAIIB CAIIB Prep

@JAIIB\_CAIIB\_Prep\_By\_MaheshSir · 6.35K subscribers · 645 videos

Welcome to JAIIB CAIIB Prep ...more

Customize channel

Manage videos

Home

Videos

Shorts

Live

Playlists

Community



CAIIB BFM Marathon Session | BFM

6,075 views · 2 months ago

#CAIIB2024 #MaheshSir #CAIIBBF  
CAIIB BFM Marathon Session | BFM  
video | CAIIB BFM Concept Classes  
Telegram - <https://t.me/jaiibcaiibpr>  
WhatsApp - <https://whatsapp.com/>  
App Link - <https://edushield.page>  
BFM Booster Batch: <https://ecxoio>  
READ MORE

JAIIB Exam 2024 | AFM | Macmillan Summary | By Mahesh Sir ▶ Play all



JAIIB Scorecard 2024 | Salary : Increment After Classing : JAIIB Exam 2024 : JAIIB Oct 2024 | JAIIB Oct 2024 | JAIIB Oct 2024 | JAIIB Oct 2024

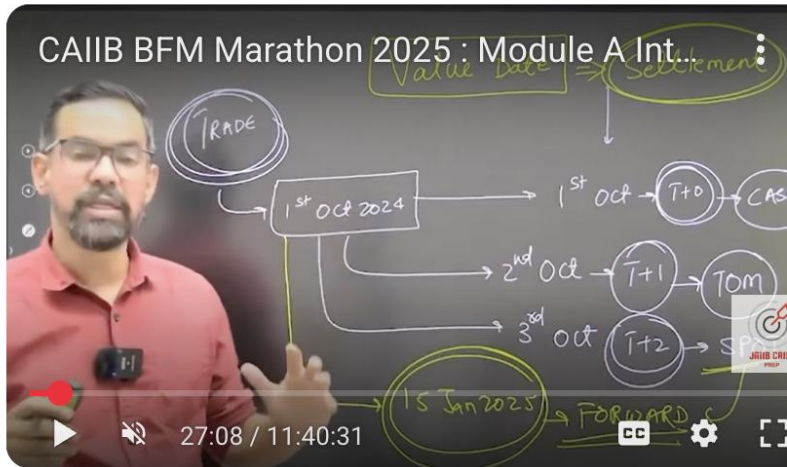
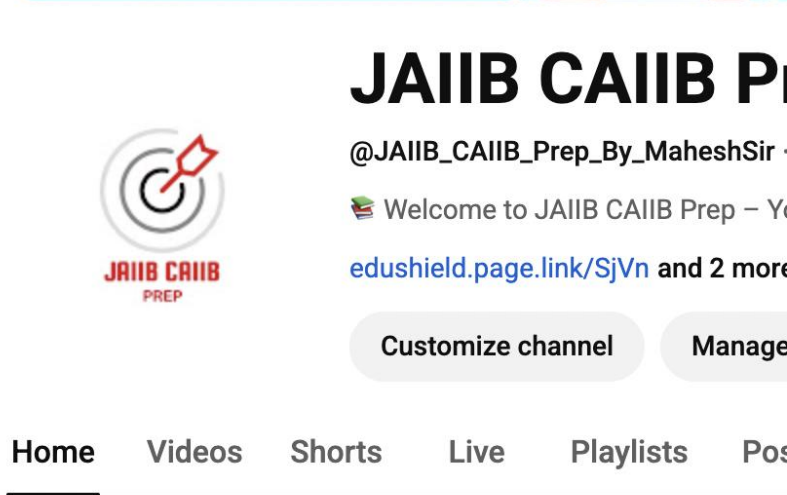


**@JAIIBCAIIBPREPOFFIC**  
**IAL**







@JAIIBCAIIBPREPOFFICIAL

<https://www.jaiibcaiibprep.com/>

# MODULE C



## TREASURY MANAGEMENT

- Unit 19: Introduction to Treasury Management
- Unit 20: Treasury Products
- Unit 21: International Equity and Debt Products
- Unit 22: Funding and Regulatory Aspects
- Unit 23: Treasury Risk Management
- Unit 24: Derivative Products
- Unit 25: Treasury and Asset-Liability Management





@JAIIBCAIIBPREPOFFICIAL

# UNIT 19

## Introduction to Treasury Management

### STRUCTURE

- 19.0 Objectives
- 19.1 The Concept
- 19.2 Functions of Integrated Treasury
- 19.3 The Process of Globalisation
- 19.4 Evolving Role of Treasury as Profit Centre
- 19.5 Organisation of Treasury

*Let Us Sum Up*

*Keywords*

# Treasury Management

The Treasury function was confined to funds management - maintaining adequate cash balances to meet day-to-day requirements, deploying surplus funds generated from operations, and sourcing funds to bridge occasional gaps in cash flow.

In the context of a bank, the Treasury is also **responsible to meet the reserve requirements**, viz. holding minimum cash balances required as per Cash Reserve Ratio (CRR) with Reserve Bank of India, and investing funds in approved securities to the extent required under Statutory Liquidity Ratio (SLR).

Thus, the Treasury function was essentially liquidity management, and from an organizational point of view, Treasury was considered as a service centre.

# Integrated Treasury

**Integrated Treasury, in a banking set-up, refers to integration of money market, securities market and foreign exchange operations.**

Integrated Treasury, in the Indian context, is the direct result of reforms in the financial sector, the most important reforms being deregulation of interest rates and partial convertibility of Rupee.

Rupee is already freely convertible on current account, and to a large extent, also convertible on capital account, owing to major relaxations allowed by the Reserve Bank, in the area of foreign direct investment (FDI), external commercial borrowings (ECB) and overseas direct investment (ODI) by Indian corporates, and foreign currency operations of resident Indians.

# Driving force Integrated Treasury

- **Integrated Cash Flow Management**
- **Interest Arbitrage**
- **Access to global resources**
- **Corporate demand for high-end services, and**
- **Risk Management**
- **Meeting reserve requirements (CRR and SLR)**
- **Global cash management**
- **Efficient merchant services, which include foreign exchange (forex) and advisory services**



# Functions of Integrated Treasury

➤ Optimising profit by exploiting market opportunities in

i)Forex market

ii)Money market

iii)Securities market (debt, equity and credit derivative markets)

➤ Risk management, i.e., managing the market risk of the  
bank/entity

➤ Assisting bank management in ALM

# Role of Treasury Management

- **Liquidity Management**
- **Proprietary Positions**
- **Risk Management:**



# Globalisation

**Globalisation refers to interaction between domestic and global markets, and may generally be defined as:**

***The process of integrating domestic market with global markets, characterised by free capital flows and minimum regulatory intervention.***

- **Portfolio investment:** Foreign investors investing in domestic equity and debt markets
- **Direct investment:** Foreign companies and foreign institutional investors investing long term funds in domestic companies, new projects, manufacturing facilities, business process outsourcing etc.
- External commercial borrowings
- Issue of equity/debt in global markets
- **Mergers & acquisitions** - involving domestic and foreign entities
- Payment for technology transfer, royalties, financial services etc

# Organisation of Treasury

**The Treasury is organised either as a Department of the bank, or as a Specialised Branch under direct control of the bank's head office.**

**Treasury will maintain an ALM Book, a Merchant Book and a trading Book, corresponding to its role in balance sheet management, merchant services and trading, across the currency and security markets.**

- **The Dealing Room (or, Front Office),**
- **The Back Office (or, Treasury Administration)**
- **The Mid-office (Risk Management).**

**The dealing room** is headed by Chief Dealer, who is in charge of the front office. The Dealers working under him, buy and sell in the markets. Each Dealer specializes in one of the markets, i.e., foreign exchange, money market or securities market, although, in an integrated treasury, the dealers are generally familiar with all the markets. Depending on the size of operations, there may be dealers dedicated to major currencies, or dealers specializing only in forward markets or derivatives. It is also common to have a separate corporate dealer, exclusively to attend to major corporate customers/ merchant business. In larger banks, Treasury will also have an ALM desk, to exclusively manage Bank's ALM risks.

**The back-office** is responsible for verification and settlement of the deals concluded by the dealers. The deals are verified on the basis of deal slips prepared by the dealers and also from the confirmation received from the counterparties. The back-office confirms the deals independently with the counterparties (banks and other institutions) over phone or digitally and verifies the authenticity of the confirmation document. The Back- office takes care of all related book-keeping and submission of periodical returns to RBI.

**Middle office (Mid-office)** is created exclusively to provide information to the management (MIS) and to implement risk management systems. Mid-office monitors exposure limits and stop loss limits of Treasury and reports to the management on key parameters of performance. Transfer Pricing Mechanism may also be implemented through Mid-office. In smaller banks, Mid-office may also function as ALM Support Group, as the balance sheet risk management is closely connected to Treasury risk management.

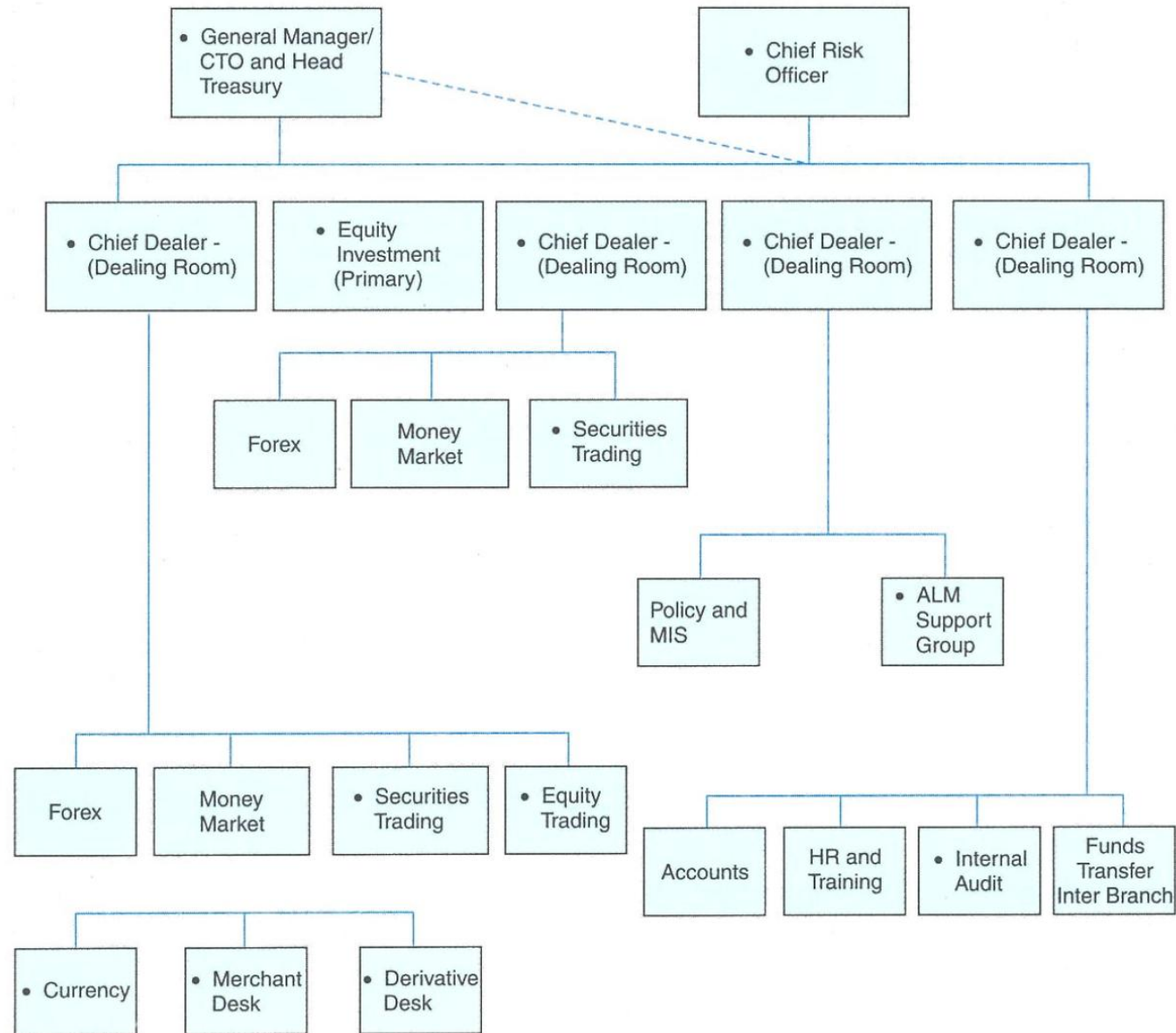


FIGURE 19.1

Treasury Organisational Structure of a Bank



# UNIT 20

## Treasury Products

### STRUCTURE

20.0 Objectives

20.1 Products of Foreign Exchange Markets

20.2 Money Market Products

20.3 Securities Market Products

20.4 Domestic and Global Markets

*Let Us Sum Up*

*Keywords*

# Treasury Management

**The Treasury function was confined to funds management - maintaining adequate cash balances to meet day-to-day requirements, deploying surplus funds generated from operations, and sourcing funds to bridge occasional gaps in cash flow.**

**Foreign exchange (forex) market is the most liquid market as free currencies (major currencies which are fully convertible, e.g. USD, EUR, GBP, JPY, CHF, etc.) can be readily bought and sold here.**

# Products of FOREX Market

Free currencies belong to those countries, whose **markets are highly developed and where exchange controls are practically dispensed.**

Currencies which are not fully convertible, **have limited demand and may not be traded actively**, but they may also enjoy high liquidity, depending on the size and stage of development of domestic market.

**For instance, Indian Rupee (INR) is only partially convertible, but the market for USD/INR is fairly liquid, owing to large domestic market and high growth rate of the economy.**

# Spot Market

**Currencies are mostly bought and sold in spot trades. The spot refers to settlement - payment and receipt of funds in respective currencies.**

**Spot settlement takes place two working days from the trade date i.e. on the third day.**

**Currency may also be bought and sold, with settlement on the same day, i.e. today (TOD/Cash, ac, on the next day, i.e. tomorrow (TOM)).**

All the exchange rates quoted on the screen, or in print, are for spot trade by default, unless otherwise mentioned.

The TOD and TOM rates are generally quoted at a discount to the spot rate, ie. the rate is less favourable to the buyer of the currency.

# Forward Market

While spot trade refers to current transaction, **forwards refer to purchase or sale of a currency on a future date. The** exchange rates for forward sale or forward purchase are quoted today; hence such transactions are referred to as forward contracts between the buyer and seller.

Treasury may enter into forward contracts with customers (merchant business) or with banks (inter-bank market) as counterparties. Customers, i.e. importers, exporters and others who expect payments or receipts in foreign currency, cover their currency risk by entering into forward contracts with their respective banks.

# Swaps

**The spot and forward transactions are the primary products in foreign exchange market. A combination of spot and forward transactions, or a combination of two forward transactions is called a swap.**

A swap transaction is also described as an exchange of cash flows. Buying USD (with Rupees) in the spot market and selling the same amount of USD in the forward market, or vice versa, constitutes a USD/INR swap. Similarly, simultaneous purchase and sale of currency on two forward dates (forward to forward) is also a swap.

# Money Market Instrument

**Money Markets refer to raising and deploying short-term resources, with maturity of funds generally not exceeding one year.**

The inter-bank market is sub-divided into call money, notice money and term money market.

| <i>Sr. No.</i> | <i>Participant Category</i>                                   | <i>Prudential Limit</i>                                                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1              | Scheduled Commercial Banks<br>(including Small Finance Banks) | <p><b>Call and Notice Money:</b><br/>           (i) 100% of capital funds, on a daily average basis in a reporting fortnight, and<br/>           (ii) 125% of capital funds on any given day.</p> <p><b>Term Money:</b><br/>           Internal board approved limit within the prudential limits for inter-bank liabilities.</p> |
| 2              | Payment Banks, and Regional Rural Banks                       | <p><b>Call, Notice and Term Money:</b><br/>           (i) 100% of capital funds, on a daily average basis in a reporting fortnight, and<br/>           (ii) 125% of capital funds on any given day.</p>                                                                                                                           |
| 3              | Co-operative Banks                                            | <p><b>Call, Notice and Term Money:</b><br/>           2.0% of aggregate deposits as at the end of the previous financial year.</p>                                                                                                                                                                                                |
| 4              | Primary Dealers                                               | <p><b>Call and Notice Money:</b><br/>           225% of Net Owned Fund (NOF) as at the end of the previous financial year on a daily average basis in a reporting fortnight.</p> <p><b>Term Money:</b><br/>           225% of Net Owned Fund (NOF) as at the end of previous financial year.</p>                                  |

# Domestic & Global Markets

**FII Investments:** Foreign investments flow in to India by way of foreign direct investment (long term project related investments), and portfolio investments (investment in stock market and debt market for short term gains). Foreign institutional investors who include investment banks and hedge funds invest mainly in portfolio investments.

**ADR GDR Issues by Indian Companies:** Indian companies mobilize foreign currency resources by issue of equity in global markets. The holders of ADR/GDR (foreign investors) have an option to sell their holdings in domestic markets, and receive proceeds in foreign currency.

**External Commercial Borrowings (ECB):** Indian companies can borrow in global markets, from banks (syndicated loans) or issue debt paper (floating rate notes, bonds etc.) within the guidelines issued by RBI, to fund their domestic/overseas projects.

**Foreign Currency Funds of Banks:** Banks can use their FCNR deposit funds for investment in overseas markets as well as for domestic lending in foreign currency. They are also permitted to borrow/invest in overseas markets within a ceiling (presently 100% of Unimpaired Tier-1 capital, minimum USD 10 million), subject to guidelines issued by RBI. Banks generally use this facility to extend short term loans (usually with 6-month rollover).

**Special Facilities to Exporters:** Exporters are permitted to hold export earnings in foreign currency accounts, designated as EEFC accounts. Banks are allowed to extend pre-shipment and post-shipment finance to exporters in foreign currency (PCFC and PSFC, respectively) in four major currencies. Banks can borrow foreign currency for the specific purpose of financing export credit (including discount of export bills).

# Domestic & Global Markets

**Free remittance Facility:** Individuals are now permitted to remit overseas freely, without RBI approval, up to USD 2,50,000 in a financial year under RBI's Liberalized Remittance Facility/Remittance, for any purpose (with a few exceptions like gambling and margin trading). They may choose to invest the funds in global debt, equity or simply spend the money for consumption purposes.

# Securities Market Products

**Investment Business** is an important part of integrated treasury and is composed of buying and selling products available in Securities Market.

Investment being a subject in itself, we would outline briefly the investment products available to bank treasuries and the subject is to be studied in detail by those who would like to specialise in investment business.

**Government Securities** are issued by Public Debt Office of Reserve Bank of India on behalf of Government of India. State governments also issue State Development Loans (in the form of Bonds) through RBI.

Government Securities are sold through auctions conducted by RBI. The interest is paid on the face value of the bonds, generally ₹100 (expressed as percentage; minimum value of bonds is Rs. 10,000) at coupon rate, but the price of the bonds is determined in the auction conducted by RBI.

# Corporate Debt Paper

Yields on corporate debt differs from instrument to instrument, depending on the credit quality, that is, higher the credit risk, higher is the yield. Most of the debt issues have credit rating by the credit rating agencies registered with SEBI. **These are:**

- The Credit Rating Information Services of India Limited (CRISIL)
- ICRA Limited
- Credit Analysis & Research Limited (CARE)
- Fitch Ratings India Private Limited
- Acuite Ratings & Research Limited
- Infomerics Valuation and Rating Private Limited (IVRPL) Global ratings are necessary if the debt paper is being issued

## Debenture and Bonds

These are debt instruments, issued by corporate bodies, literally with a charge on specific assets. The literal meaning has been lost in practice and debentures and bonds may be issued with or without security. In practice, Company Law requires that debentures issued by companies are always secured (otherwise they will be subject to regulations pertaining to public deposits); hence debentures are generally secured by mortgage or with a floating charge or a lien on assets – although the latter security is more of a technical nature to give comfort to the investors.

## Equity

Equity share capital can be:

- With voting rights; or
- With differential rights as to dividend, voting or otherwise

Preference share capital is that capital which enjoys preferential right with respect to the dividend payment at a fixed rate and return of the capital on winding up of the company.



## **Preference Shares can be further classified as:**

**Cumulative or non-cumulative** - rights to claim dividend does not lapse when there is no profit or inadequate profit and is carried forward to the next year is cumulative preference shares while rights to claim dividend lapses in case of non-cumulative shares.

**Redeemable or irredeemable** - which can be redeemed, subject to fulfilment of certain conditions laid down in Section 55 of the Companies Act, 2013. Section 55(i) of the Companies Act 2013 prohibits the issue of any preference share that is irredeemable.

**Participating or non-participating** - which have a right to participate in the surplus remaining after payment of equity capital in case of winding up of the company.

**Convertible or non-convertible** - which would be convertible into or exchanged with equity shares at a later date, with or without the option of the preference shareholders



<https://www.jaiibcaiibprep.com/>

## UNIT 21

# International Equity and Debt Products

### STRUCTURE

- 21.0 Objectives
- 21.1 Regulatory Environment
- 21.2 Global Depository Receipts (GDRs)
- 21.3 Indian Depository Receipts (IDRs)
- 21.4 External Commercial Borrowings
- 21.5 Trade Credits
- 21.6 Rupee Denominated Bonds

*Let Us Sum Up*

*Keywords*



# Regulatory Environment

**Foreign Investment in India is regulated in terms of clause (b) sub-section 3 of section 6 and section 47 of the Foreign Exchange Management Act, 1999 (FEMA) read with Foreign Exchange Management (Transfer or Issue of a Security by a Person resident Outside India) Regulations, 2017 issued vide Notification No. FEMA 20(R)/2017-RB dated November 7, 2017.**

‘Foreign Direct Investment’ (FDI) is the investment through capital instruments by a person resident outside India (a) in an unlisted Indian company; or (b) in 10 percent or more of the post issue paid-up equity capital on a fully diluted basis of a listed Indian company.

# Prohibited Sectors/Persons

*Investment by a person resident outside India is prohibited in the following sectors:*

- Lottery Business including Government/ private lottery, online lotteries.
- Gambling and betting including casinos.
- Chit funds (except for investment made by NRIs and OCIs on a non-repatriation basis).
- Nidhi company.
- Trading in Transferable Development Rights (TDRs).
- Manufacturing of Cigars, cheroots, cigarillos and cigarettes, of tobacco or of tobacco substitutes.
- The prohibition is on manufacturing of the products mentioned and foreign investment in other activities relating to these products including wholesale cash and carry, retail trading etc. will be governed by the sectoral restrictions laid down in Regulation 16 of FEMA 20(R).
- Activities/ sectors not open to private sector investment viz.,
  - (i) Atomic energy and (ii) Railway operations.

# Regulatory Environment

‘Foreign Portfolio Investment’ is any investment made by a person resident outside India in capital instruments where such investment is (a) less than 10 percent of the post issue paid-up equity capital on a fully diluted basis of a listed Indian company or (b) less than 10 percent of the paid up value of each series of capital instruments of a listed Indian company.

‘Foreign Portfolio Investor (FPI)’ is a person registered in accordance with the provisions of Securities Exchange Board of India (Foreign Portfolio Investors) Regulations, 2014.

# External Commercial Borrowing (ECB)

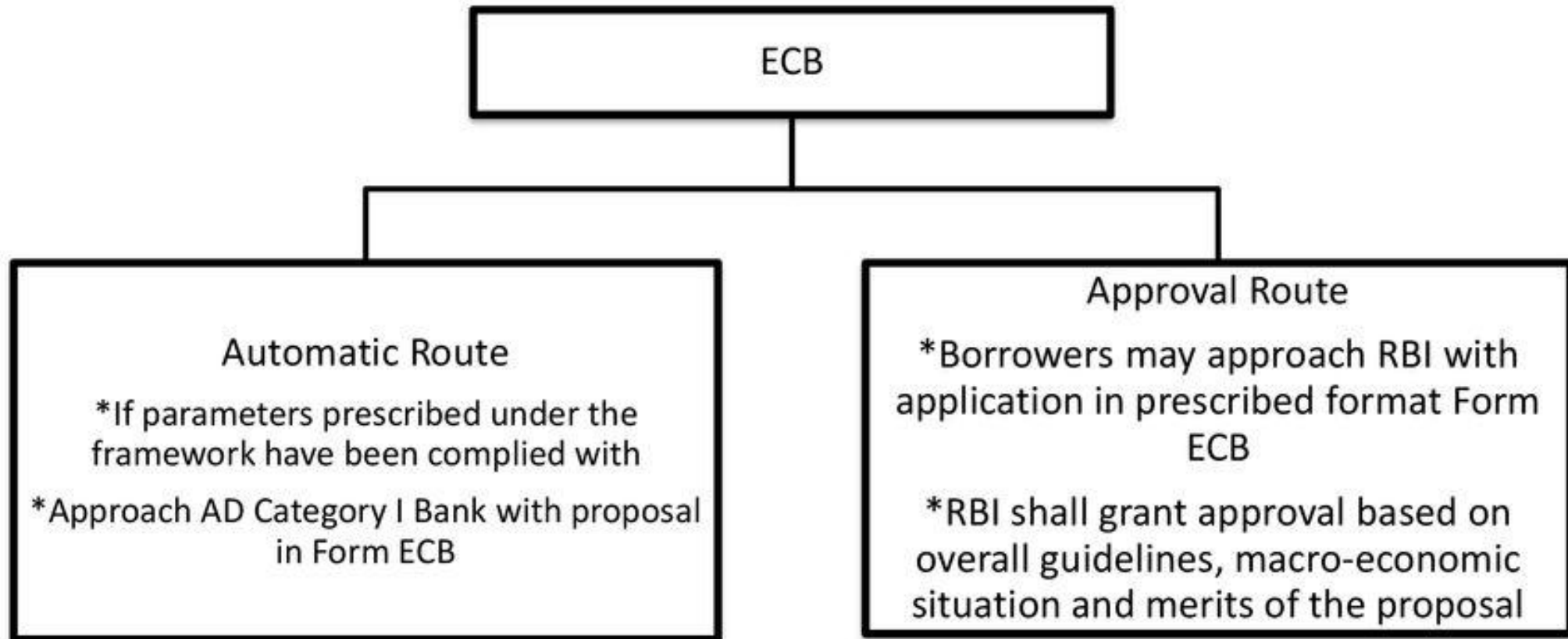
External Commercial Borrowing (ECB) refers to the **borrowing of funds by Indian companies from foreign sources** in the form of loans, bonds, or other financial instruments.

ECB can be used to finance a variety of purposes, including the expansion of business, the acquisition of assets, and the repayment of existing debt.

ECB is subject to regulatory oversight by **the Reserve Bank of India (RBI)**, which sets limits on the amount of ECB that Indian companies can obtain and the purposes for which it can be used. Companies must also meet certain eligibility criteria in order to access ECB, such as minimum credit ratings and debt-equity ratios.



# Procedure of raising ECB

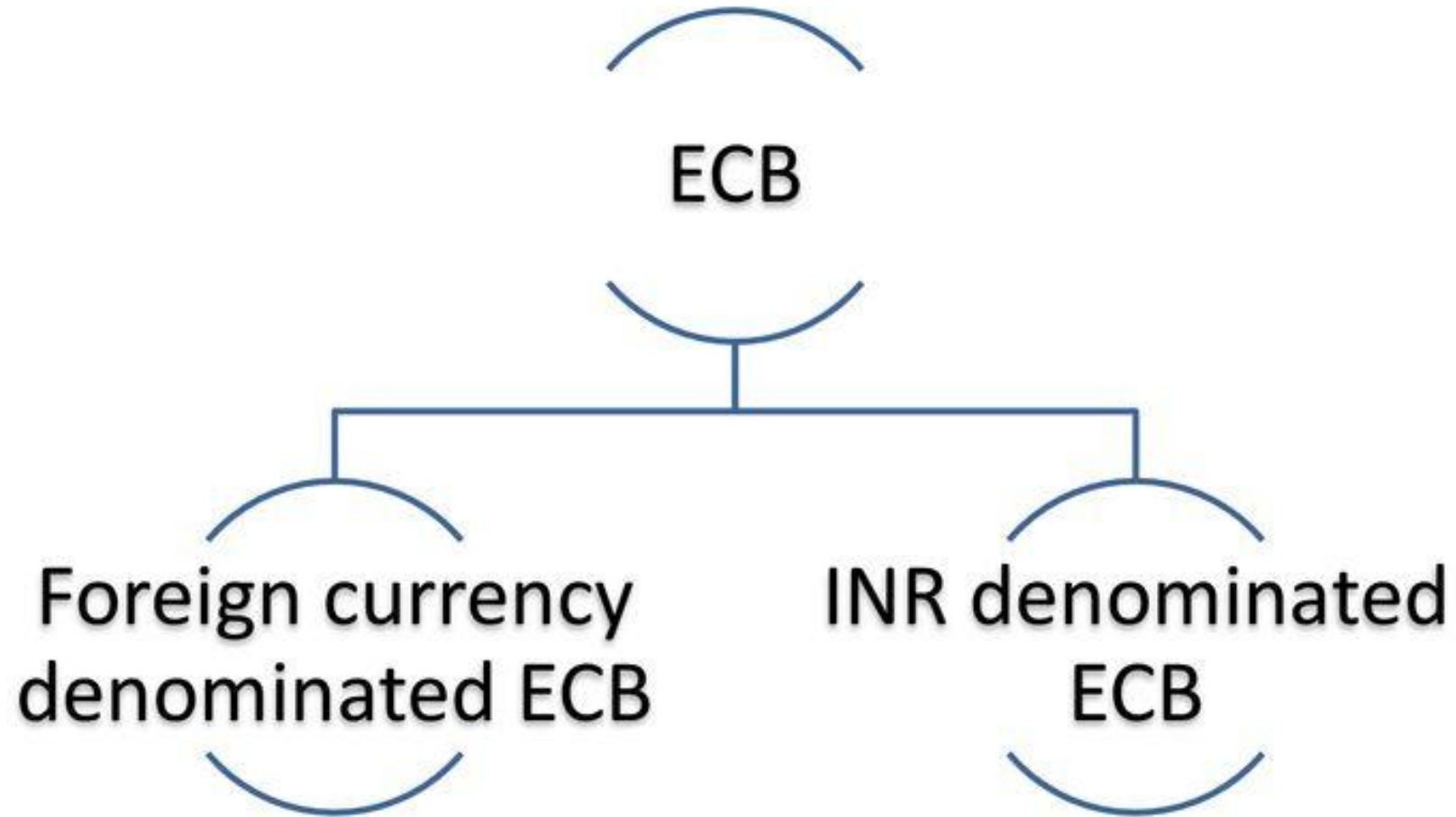


# External Commercial Borrowing (ECB)

The eligible entities can raise funds through ECB either under **the Automatic route where the cases will be examined by the AD I** or through the Approval route, under which prospective **borrowers are require to obtain RBI approval** by routing their request through their AD Banks for examination.



# Forms of ECB



# ECB

| <i>Description</i> | <i>ECB in Foreign Currency</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>ECB in Indian Rupee</i>                                                                                                                                                                                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Currency           | Any freely convertible Foreign Currency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Indian Rupee                                                                                                                                                                                                                                                                                                                                                          |
| Instrument Type    | Loans including bank loans; floating/ fixed rate notes/ bonds/ debentures (other than fully and compulsorily convertible instruments); Trade credits beyond 3 years; FCCBs; FCEBs and Financial Lease.                                                                                                                                                                                                                                                                                                                                                                                                   | Loans including bank loans; floating/ fixed rate notes/bonds/ debentures/ preference shares (other than fully and compulsorily convertible instruments); Trade credits beyond 3 years; and Financial Lease. Also, plain vanilla Rupee denominated bonds issued overseas, which can be either placed privately or listed on exchanges as per host country regulations. |
| Eligible Borrower  | All entities eligible to receive FDI. Further, the following entities are also eligible to raise ECB:<br>i. Port Trusts;<br>ii. Units in SEZ;<br>iii. SIDBI; and<br>iv. EXIM Bank of India.                                                                                                                                                                                                                                                                                                                                                                                                              | a) All entities eligible to raise FCY ECB; and<br>b) Registered entities engaged in micro-finance activities, viz., registered Not for Profit companies, registered societies/ trusts/ cooperatives and Non-Government Organisations.                                                                                                                                 |
| Recognised Lenders | The lender should be resident of FATF or IOSCO compliant country, including on transfer of ECB. However,<br>a) Multilateral and Regional Financial Institutions where India is a member country will also be considered as recognised lenders;<br>b) Individuals as lenders can only be permitted if they are foreign equity holders or for subscription to bonds/debentures listed abroad; and c) Foreign branches / subsidiaries of Indian banks are permitted as recognised lenders only for FCY ECB (except FCCBs and FCEBs). Foreign branches / subsidiaries of Indian banks, subject to applicable |                                                                                                                                                                                                                                                                                                                                                                       |

# MAMP

Minimum Average  
Maturity Period  
(MAMP)

MAMP for ECB will be 3 years. Call and put options, if any, shall not be exercisable prior to completion of minimum average maturity. However, for the specific categories mentioned below, the MAMP will be as prescribed therein:

| Sr.No.           | Category                                                                                                                                                            | MAMP     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| (a)              | ECB raised by manufacturing companies up to USD 50 million or its equivalent per financial year.                                                                    | 1 year   |
| (b)              | ECB raised from foreign equity holder for working capital purposes, general corporate purposes or for repayment of Rupee loans.                                     | 5 years  |
| <sup>1</sup> (c) | ECB raised for<br>(i) working capital purposes or general corporate purposes<br>(ii) on-lending by NBFCs for working capital purposes or general corporate purposes | 10 years |
| (d)              | ECB raised for<br>(i) repayment of Rupee loans availed domestically for capital expenditure<br>(ii) on-lending by NBFCs for the same purpose                        | 7 years  |
| (e)              | ECB raised for<br>(i) repayment of Rupee loans availed domestically for purposes other than capital expenditure<br>(ii) on-lending by NBFCs for the same purpose    | 10 years |

For the categories mentioned at (b) to (e) –

- (i) ECB cannot be raised from foreign branches / subsidiaries of Indian banks
- (ii) the prescribed MAMP will have to be strictly complied with under all circumstances.

# ECB

| <i>Description</i>              | <i>ECB in Foreign Currency</i>                                                                                                                                                                                                                              | <i>ECB in Indian Rupee</i>                                                                      |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                                 | ECB may be considered naturally hedged if the offsetting exposure has the maturity/cash flow within the same accounting year. Any other arrangements/ structures, where revenues are indexed to foreign currency will not be considered as a natural hedge. |                                                                                                 |
| Change of Currency of Borrowing | Change of currency of ECB from one freely convertible foreign currency to any other freely convertible foreign currency as well as to INR is freely permitted.                                                                                              | Change of currency from INR to any freely convertible foreign currency is <b>not</b> permitted. |

# ECB \_\_ Eligible Borrowers

- All entities eligible to receive FDI
- Entities recognized as Start-up by the Central Government as on date of raising ECB.
- Port Trust, Units in SEZ, SIDBI, EXIM Bank
- Registered entities engage in micro-finance activities

# ECB \_\_ End Use

ECB Proceeds cannot be utilized would include the following

- Investment in capital market
- Equity investment
- Working capital purposes except from foreign equity holder
- General corporate purposes except from foreign equity holder
- Repayment of Rupee loans except from foreign equity holder
- On-lending to entities for the above activities



# ECB \_\_ Eligible Borrowers

- A)** Borrowers can raise ECB from internationally recognised sources such as (i) international banks, (ii) international capital markets, (iii) multilateral financial institutions (such as IFC, ADB, CDC etc.), (iv) export credit agencies, (v) suppliers of equipment, (vi) foreign collaborators and (vii) foreign equity holders (other than erstwhile OCBs).
- B)** Overseas organisations and individuals complying with following safeguards may provide ECB to Non-Government Organisations (NGOs) engaged in micro finance activities.
- Overseas organisations** proposing to lend ECB would have to furnish a certificate of due diligence from an overseas bank which in turn is subject to regulation of host-country regulator and adheres to **Financial Action Task Force (FATF)** guidelines to the AD bank of the borrower.

# Global Depository Receipts (GDR)

GDRs represent Receipts that entitle the holder **to convert into specified number of equity shares of Indian Company**. The Receipts are issued by a Depository abroad and are traded in overseas markets. They are negotiable Receipts.

**The underlying shares, issued by the Indian Company**, are held by an Indian Custodian on behalf of the Overseas Depository. They are denominated in foreign currency.

# Global Depository Receipts (GDR)

- ✓ The exchange risk on the GDR is borne by the overseas investor. The equivalent number of equity shares is fixed as per pricing norms of SEBI. On issuance of GDR, the equity of the issuing company increases.
- ✓ Dividend is paid out in Rupees to the Depository.
- ✓ The Depository is entitled to voting rights as it holds the equity shares on behalf of the GDR holders.
- A limited two way fungibility scheme is in operation by Government of India for ADRs/ GDRs.
- Under this, a SEBI registered Stock Broker can purchase the shares from the market for conversion into ADRs/ GDRs.
- Reissuance of ADR/GDR would be permitted to the extent of ADRs/GDRs that have been redeemed into underlying shares and sold in the domestic market.

# Indian Depository Receipts (IDR)

- ✓ Issue of IDRs Companies incorporated outside India may issue IDRs through a Domestic Depository, to a person resident in India and a person resident outside India. The issue of IDRs should comply with the Companies (Registration of Foreign Companies) Rules, 2014 and the Securities and Exchange Board of India (Issue of Capital and Disclosure Requirements) Regulations, 2009;
- ✓ Any issue of IDRs by financial/ banking companies having presence in India, either through a branch or subsidiary, shall require prior approval of the sectoral regulator(s);
- ✓ IDRs shall be denominated in Indian Rupees only;
- ✓ The proceeds of the issue of IDRs shall be immediately repatriated outside India by the companies issuing such IDRs. Purchase/Sale of IDRs An FPI or an NRI or an OCI may purchase, hold or sell IDRs NRIs or OCIs may invest in the IDRs out of funds held in their NRE/ FCNR(B) account, maintained in accordance with the Foreign Exchange Management (Deposit) Regulations, 2016. There would be an overall cap of USD 5 billion for raising of capital by issuance of IDRs by eligible foreign companies in Indian markets. This limit would be monitored by SEBI.

**Investment by a person resident outside India is prohibited in the following sectors:**

- 1) Lottery Business
- 2) Gambling and betting
- 3) Chit funds (except for investment made by NRIs and OCIs on a non-repatriation basis)
- 4) Nidhi company
- 5) Trading in Transferable Development Rights (TDRs)
- 6) Real estate business of construction of farm houses
- 7) Manufacturing of cigars, cheroots, cigarillos and cigarettes, of tobacco or tobacco substitutes.
- 8) Activities/ sectors not open to private sector investment viz, atomic energy and railway operations
- 9) Foreign technology collaboration in any form including licensing for franchise, trademark, brand name, is also prohibited for Lottery Business and Gambling and betting activities.

**The negative list for which the ECB proceeds cannot be utilised, would include the following:**

- 1) Real estate activities
- 2) Investment in capital market
- 3) Equity investment
- 4) Working capital purpose, except in case of ECB mentioned at (2) and (3) above
- 5) General corporate purposes, except in case of ECB mentioned at (2) and (3) above
- 6) Repayment of Rupee loans, except in case of ECB mentioned at (4) and (5) above
- 7) On-lending to entities for the above activities, except in case of ECB raised by NBFC as given at (3), (4) and (5) above.

# Trade Credit

| Sr. No. | Parameters                    | FCY denominated TC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INR denominated TC                                                                                                                                                   |
|---------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i       | Forms of TC                   | Buyers' Credit and Suppliers' Credit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                      |
| ii      | Eligible borrower             | Person resident in India acting as an importer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                      |
| iii     | Amount under automatic route  | Up to USD 150 million or equivalent per import transaction for oil/gas refining & marketing, airline and shipping companies. For others, up to USD 50 million or equivalent per import transaction.                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                      |
| iv      | Recognised lenders            | 1. For suppliers' credit: Supplier of goods located outside India.<br>2. For buyers' credit: Banks, financial institutions, foreign equity holder(s) located outside India and financial institutions in IFSCs located in India.<br>Note: Participation of Indian banks and non-banking financial companies (operating from IFSCs) as lenders will be subject to the prudential guidelines issued by the concerned regulatory departments of the Reserve Bank. Further, foreign branches/subsidiaries of Indian banks are permitted as recognised lenders only for FCY TC. |                                                                                                                                                                      |
| v       | Period of TC                  | The period of TC, reckoned from the date of shipment, shall be up to three years for import of capital goods. For non-capital goods, this period shall be up to one year or the operating cycle whichever is less. For shipyards / shipbuilders, the period of TC for import of non-capital goods can be up to three years.                                                                                                                                                                                                                                                |                                                                                                                                                                      |
| vi      | All-in-cost ceiling per annum | <sup>12</sup> Benchmark Rate plus 350 bps spread: For existing TCs linked to LIBOR whose benchmarks are changed to ARR.<br>Benchmark rate plus 300 bps spread: For new TCs.                                                                                                                                                                                                                                                                                                                                                                                                | Benchmark rate plus 250 bps spread.                                                                                                                                  |
| vii     | Exchange rate                 | Change of currency of FCY TC into INR TC can be at the exchange rate prevailing on the date of the agreement between the parties concerned for such change or at an exchange rate, which is less than the rate prevailing on the date of agreement, if consented to by the TC lender.                                                                                                                                                                                                                                                                                      | For conversion to Rupee, exchange rate shall be the rate prevailing on the date of settlement.                                                                       |
| viii    | Hedging provision             | The entities raising TC are required to follow the guidelines for hedging, if any, issued by the concerned sectoral or prudential regulator in respect of foreign currency exposure. Such                                                                                                                                                                                                                                                                                                                                                                                  | The overseas investors are eligible to hedge their exposure in Rupee through permitted derivative products with AD Category I banks in India. The investors can also |



@JAIIBCAIIBPREPOFFICIAL

<https://www.jaiibcaiibprep.com/>

# UNIT 22

## Funding and Regulatory Aspects

### STRUCTURE

- 22.0 Objectives
- 22.1 Reserve Assets: CRR and SLR
- 22.2 The Liquidity Adjustment Facility (LAF)
- 22.3 Payment and Settlement Systems

*Let Us Sum Up*

*Keywords*

*Multiple Choice Questions*

*Answers to the MCQs*



## Reserve Assets : CRR & SLR

The Reserve Bank of India is the Note Issuing Authority, that is, the currency in circulation is directly controlled by RBI. However, the currency is only cash component of money in circulation, and in that, it is only a small part of the total money. **The cash deposited in banks in turn is lent by the banks, which increases supply of money.**

If part of the money, so borrowed, is held in a deposit account with the bank, the chain of relending and creating new deposits will continue. It is not only cash, but near cash instruments like cheques and credit cards also add to the money supply (e.g. money spent on credit card is deposited with a bank, adding further money). Creation of money in **this fashion is called the money multiplier effect.**

# Main components of DTL

- Demand deposits (held in current and savings accounts, margin money for LCs, overdue fixed deposits etc.).
- Time deposits in fixed deposits, recurring deposits, reinvestment deposits etc.).
- Overseas borrowings.
- Foreign outward remittances in transit (FC liabilities net of FC assets).
- Other demand and time liabilities (accrued interest, credit balances in suspense account etc.).

## **Scheduled Commercial Banks are exempted from maintaining CRR on the following liabilities:**

- Liabilities to the banking system in India as computed under clause (d) of the explanation to Section 42(1) of the RBI Act, 1934.
- Credit balances in ACU (US\$) Accounts.
- Demand and Time Liabilities in respect of their Offshore Banking Units (OBUs).
- The eligible amount of incremental FCNR (B) and NRE deposits of maturities of three years and above from the base date of July 26, 2013, and outstanding as on
- March 7, 2014, till their maturities/ pre-mature withdrawals.
- Minimum of Eligible Credit (EC) and outstanding Long term Bonds (LB) to finance Infrastructure Loans and affordable housing loans.

## The following liabilities are excluded from the CRR stipulation:

- Paid-up capital, reserves, retained profits, refinance availed from RBI, and apex financial institutions like NABARD and SIDBI.
- Net income tax provision.
- Claims received from DICGC, ECGC, Insurance Company (Towards ad-hoc settlement), Court Receiver etc.
- Liabilities arising on account of utilization of limits under Bankers Acceptance Facility.
- District Rural Development Agency (DRDA) subsidy of Rs. 10,000/- kept in Subsidy Reserve Fund account in the name of Self Help Groups.
- Subsidy released by NABARD under Investment Subsidy Scheme for Construction/Renovation/ Expansion of Rural Godowns .

**The SLR is to be maintained in the form of the following assets:**

- Cash balances (excluding balances maintained for CRR).
- Gold (valued at price not exceeding current market price).
- Approved securities valued as per norms prescribed by RBI,

## Liquidity Adjustment Facility (LAF)

**The Liquidity Adjustment Facility is the principal operating instrument of Reserve bank's monetary policy.** While CRR and SLR help changes in the stance of monetary policy on a more permanent basis LAF is used to monitor day-to-day liquidity in the market.

LAF refers to RBI lending funds to banking sector through Repo instrument. RBI also accepts deposits from banks under Reverse Repo. The process of purchase and sale of government securities, with agreement to sell-back or repurchase respectively, within a predetermined period .

## Liquidity Adjustment Facility (LAF)

While banks can engage in repo transactions with other banks/institution, LAF refers exclusively to repo transactions with RBI. **Bids have to be submitted for a minimum amount of Rs. 5 crore and in multiples of Rs. 5 crore thereafter.**

## Liquidity Adjustment Facility (LAF)

***In order to help banks, to wade through the liquidity constraints, RBI has taken the following measures on February, 3, 2015:***

- Continue to provide liquidity under overnight repos (fixed rate repo) of 0.25% of bank-wise NDTL at the LAF repo rate.
- Introduced a new window called Variable Rate Repo. Under this window liquidity would be provided for 7 days, 14 days and 28 days. This is also called as Term Repos. The limit fixed by RBI, is under this window, is 0.75% of NDTL of the banking system. Interested banks can avail funds under this route by quoting a rate which should be equal to or above the Repo Rate. The bank which has quoted the highest yield stands a better chance of getting funds under this window.
- RBI has also phased out the Export Credit Refinance facility.

# Payment and Settlement Systems

Payment and settlement systems play a vital role in the development of financial markets. The important reforms relevant to treasury operations include the following:

- Real Time Gross Settlement System (RTGS)
- Negotiated Dealing System
- FX Clear
- Depository Institutions
- NEFT and on-line Payments

# Negotiated Dealing System

**Negotiated Dealing System** is an electronic platform for facilitating dealing in government securities and money market instruments. RBI had introduced the NDS in February 2002, in order to achieve a) automatic electronic reporting and settlement process b) auctions on electronic platform and c) a trading platform for trading in Government securities on a negotiated basis (telephone based trading), as well as quote-driven mechanism.



# FX Clear

**FX Clear** is a forex dealing system developed by CCIL for foreign exchange transactions (USD) INR as well as cross currencies). Currently CCIL is providing straight through processing (STP) for USD/INR, and CCIL as an intermediary settles inter-bank USD/Rupee as well as cross currency deals on net basis, so that individual banks need not exchange payments for each transaction.



# Depository Institutions

**Depository Institutions** like NSDL (National Securities Depository Ltd.) and CSDL (Central Securities Depository Ltd.) provide delivery vs. payment (DVP) for secondary market deals in equity and debt paper. The securities and funds are cleared by their respective clearing houses.

# NEFT and on-line Payments

**NEFT and on-line Payments** All inter-bank and intra-bank remittances can now be effected on the same day by electronic funds transfer using the National Electronic Funds transfer system introduced by RBI. RBI has developed Structured Financial Messaging System (SFMS) - similar to SWIFT adopted by banks for international funds transfer etc. - where interbank transfers are sorted out and cleared by National Clearing Cell of RBI.



@JAIIBCAIIBPREPOFFICIAL

# UNIT 23

## Treasury Risk Management

### STRUCTURE

- 23.0 Objectives
- 23.1 Supervision and Control of Treasury
- 23.2 Market Risk
- 23.3 Risk Measures: VaR and Duration
- 23.4 Use of Derivatives in Risk Management

*Let Us Sum Up*

<https://www.jaiibcaiibprep.com/>



## Supervision and Control Of Treasury

**Treasury risk management assumes importance for two reasons:**

- The nature of treasury activity is such that profits are generated out of market opportunities and market risk is present at every step;
  - Treasury is also responsible for balance sheet management, i.e. market risk generated by other operation departments.
- We will deal with the first aspect a little more elaborately.

## Concern for Treasury Risks

Bank management is highly sensitive to treasury risk, as the risk arises out of high leverage the treasury business enjoys. The risk of losing capital is much higher than, say, in the credit business. Bank's capacity to extend loans is limited by the resources at its command, that is, deposits and other borrowings. In case of a loan, **the risk is limited to the principal and interest**, which may be lost, fully or partly, over a period of time.

## Concern for Treasury Risks

Treasury on the other hand, has a very low funding requirement, which we call as high leverage. For instance, treasury can buy and sell foreign exchange of value Rs. 100 crore without any direct investment of funds, except for allocation of risk capital as per capital adequacy requirement of RBI. At the same time, an adverse movement of the exchange rate by Re. 1 may result in a loss of over Rs. 1 crore to the bank - which is a straight loss of capital.

## Concern for Treasury Risks

Treasury risks are primarily managed by conventional control and supervisory measures, mostly in the nature of preventive steps, which may be divided into three parts:

- Organisational Controls
- Exposure Ceiling
- Limits on trading positions and stop-loss limits

## Organisational Controls

The organisational controls refer to the checks and balances  
within the system. Treasury is basically divided into three  
parts: the front office, back office and the Mid office.

## Internal Controls

The most important of the internal controls are **position limits and stop loss limits**. The limits are imposed on the dealers who trade in foreign exchange and securities. Trading is a high risk area, vulnerable to sudden market fluctuations and the limits imposed by management are preventive measures to avoid or contain losses in adverse market conditions.

**The trading limits in the context of foreign exchange are of three kinds:**

- **limits on deal size**
- **limits on open positions and**
- **stop-loss limits.**

**All limits are expressed in absolute amounts**



**@JAIIBCAIIBPREPOFFICIAL**



## Exposure Ceiling Limits

Exposure limits are kept in place to protect the bank from credit risk/counter-party risk. Credit risk in Treasury may be split into **default risk and settlement risk**.

Default risk is typically when the bank lends in the money market (mainly to other banks), the borrowing bank may fail to repay the amount on due date. Similar risk is there in repo transactions also.

## Market Risk and Credit Risk

While describing the conventional controls above, we have frequently referred to default by a counterparty and market movements.

## Market Risk and Credit Risk

Thus market risk is a confluence of liquidity risk, interest rate risk, exchange rate risk, equity risk and commodity risk.

Considering all these factors, the Bank for International Settlements (BIS) defines market risk as "the risk that the value of on- or off-balance sheet positions will be adversely affected by movements in equity and interest rate markets, currency exchange rates and commodity prices". The market risk is also known as price risk.

*The three main components of market risk are liquidity risk, interest rate risk and currency risk.*



@JAIIBCAIIBPREPOFFICIAL

➤ **Liquidity Risk**

➤ **Interest Rate Risk**

➤ **Currency Risk**



## **Liquidity Risk**

**Liquidity risk refers to cash flow gaps which could not  
be bridged. Let us assume that the Treasurer has  
borrowed in call market and purchased a 5- year  
government security, assuming the bond prices would  
go up next day and he can sell the security with profit.**

## Interest Rate Risk

**Interest rate risk refers to rise in interest costs (of a liability) or fall in interest earnings (from assets) eroding the business profits.**

Treasury deals in financial assets, value of which is highly sensitive to interest rate movements. A steep rise in interest rates may cause a crash of bond market, eroding the value of securities held by Treasury.

## Currency Risk

Currency risk or exchange rate risk is also a manifestation of interest rate risk, although for the sake of clarity, it is identified as a component of market risk. Interest rates are influenced by factors like domestic money supply, rate of inflation, activity in debt and equity markets etc. which also influence exchange rates.

However, exchange rates are influenced more by external trade, global interest rates and capital flows. As globalization progresses, exchange rates and interest rates are increasingly influenced by similar factors, most prominent being GDP growth rate, global interest rates and capital flows.

## Risk Measures: VaR and Duration



The movement in currency prices or security prices cannot be accurately predicted and the uncertainty associated with their price movements gives rise to price risk.

At the same time the treasurer should have some idea of the inherent risks and the way they would affect his positions. This quest for risk solutions, led to two important measures of risk, known as value at risk and duration.

## Value at Risk (VaR)

VaR is a statistical measure **indicating the worst possible movement of a market rate**, over a given period of time, under normal market conditions, at a defined confidence level. For instance, an overnight VaR of 45 bps for USD/INR rate at 95% confidence level implies that there is only 5% chance of the rate worsening beyond 45 bps next day. If today's spot rate is Rs. 64.00, tomorrow the worst possible rate for exports can be assumed to be Rs. 63.55. With reasonable safety - there is only 5% chance of the rate being worse than Rs. 63.55.

## Value at Risk (VaR)

Parametric approach, based on sensitivity of various risk components. This approach is also called as Correlation approach or Variance-Co-Variance approach. For instance, say the price of a stock depends on its sensitivity to index changes, to interest rate changes and to changes in the exchange rates - all these components are built into a complex formula to arrive at the VaR of the stock.

## Value at Risk (VaR)

The second approach is based on Monte Carlo simulation, where a number of scenarios are generated at random and their impact on the subject (stock price/exchange rate etc.) is studied.

## Value at Risk (VaR)

The third approach is to use historical data to arrive at the probable loss. The historical data may simply be time series of data prevailing over a period (e.g. daily USD/INR exchange rate for last 90 days), or an index of changes in price over previous day). Progressive weights may also be assigned to the data, as more recent information has greater impact on future price movements.

## Duration

Duration is a measure widely used in investment business, though the concept of duration is applicable to all assets and liabilities, where interest rate risk is present . To understand Duration, we need to be familiar with the concept of YTM or Yield to Maturity of a bond.

## Duration

The bonds carry a coupon rate of interest which is payable on 100% value of the bond (par value, as at the time of issue).

However, the bonds may be traded at a discount ( $< \text{Rs. } 100$ ) or at a premium ( $> \text{Rs. } 100$ ), depending on the interest rate trends in the market. The traded price is based on the market rate of interest for residual period of the bond and is constantly changing in the market.

# Use of Derivatives In Risk Management

**Derivatives are financial contracts which derive their value based on an underlying market for a commodity or financial product. Derivatives are used to protect treasury transactions from market risk.**

## Use of Derivatives In Risk Management

Derivatives are also useful in managing balance sheet risk i.e. asset liability management. For instance, if an asset is highly sensitive to interest rate changes, and if we have a view that the rates are likely to rise, we can swap fixed rate of interest into floating rate by entering into an interest rate swap. Similarly exchange rate risk can be avoided by entering into a forward rate contract or option contract.

# UNIT 24

## Derivative Products



@JAIIBCAIIBPREPOFFICIAL

### STRUCTURE

- 24.0 Objectives
- 24.1 Derivatives and the Treasury
- 24.2 OTC and Exchange Traded Products
- 24.3 Forwards, Options, Futures and Swaps
- 24.4 Interest Rate and Currency Swaps
- 24.5 Developments in Indian Markets, and RBI Guidelines on Risk Exposure

<https://www.jaiibcaiibprep.com/>

*Let Us Sum Up*



| <i>OTC</i>                                                                                              | <i>Exchange Traded</i>                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OTC products are offered by banks and financial institutions (need to be authorized banks in India)     | Futures contracts are traded only on organized futures exchanges                                                                                                      |
| Contract date, amount and terms as desired by the client                                                | Size of contract is standardized, with pre-set settlement dates for specific terms (e g one month, \$1000 contract settled last Wednesday of every month against INR) |
| Price is quoted by the Bank, adding a margin to market quote                                            | Transparent pricing, based on screen-based order matching system                                                                                                      |
| Security (cash margin, charge on assets etc.), at MTM bank's discretion, based on client status         | The Exchange collects daily cash margin based on value of the contract                                                                                                |
| Counter-party risk (bank risk is present but it is a party – remote risk.)                              | No counter-party risk, as Exchange is the counter which manages the risk by margining system                                                                          |
| Settlement is mostly by physical delivery (net may settlement only in trading positions/ cancellations) | Mostly net settlement by cash (physical delivery be insisted upon in commodity futures)                                                                               |
| Mostly used for hedging underlying risk                                                                 | Mostly used for trading and speculation.                                                                                                                              |

| <i>Forward Contract</i>                                                                                                         | <i>Option Contract</i>                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The contract must be executed at contracted rate on the expiry date.                                                            | The holder has a right to exercise the option, but has no obligation.                                                                                                                                                                                                                                                                    |
| The rate is fixed at current market quote.                                                                                      | Holder may choose strike price (contracted rate) or market rate whichever is better for him.                                                                                                                                                                                                                                             |
| There is no fee payable, the quoted rate includes bank margin.                                                                  | Option premium is payable front-end.                                                                                                                                                                                                                                                                                                     |
| Forward premium is the interest rate differential of the two currencies involved.                                               | Option premium is determined by several factors, including strike price, volatility of exchange rates and interest rates.                                                                                                                                                                                                                |
| Forward contract is a simple contract for purchase or sale of currency – there are no variations.                               | Various types of options are available, and simple to complex structures, with varying elements of risk, are possible by combining purchased and written options.                                                                                                                                                                        |
| Buyers and sellers have only counter-party risk, and there is no market risk to either of them, so long as the market is liquid | <p>(1) The writer of an option (option seller) has unlimited risk, while the buyer of an option has full upside. benefit, with no risk in a plain vanilla option.</p> <p>(2) There is no market risk in plain vanilla options; however structured products may expose the holder to huge risks, which he may or may not be aware of.</p> |

Derivatives are financial contracts derived from an underlying asset, either financial or commodity-based. They can be **OTC (Over-The-Counter)** or **Exchange Traded**, with banks mostly dealing in OTC derivatives for financial markets.

### **Types of Derivatives:**

- ✓ **Forward Contracts:** OTC products used to fix currency and interest rates.
- ✓ **Options:** Provide the right (not obligation) to buy (call) or sell (put) an asset at a set price.
- ✓ **Futures:** Exchange-traded forwards with fixed volumes/dates, daily margin requirements, and definite delivery.
- ✓ **Swaps:** Used for hedging and managing interest/currency risks.

## Swaps:

- ✓ **Interest Rate Swaps (IRS):** Exchange of cash flows to shift between fixed and floating rates.
- ✓ **Currency Swaps:** Exchange cash flows in different currencies, used for hedging and interest rate arbitrage in foreign loans.

## Usage by Banks:

Banks use these derivatives for hedging currency and interest rate risks and for protecting against market volatility.

## Regulatory Context in India:

- ✓ RBI permits certain derivatives like IRS and USD/INR options under restrictions.
- ✓ Benchmarks include **SOFR (ARR)**, Fed rates, MIBOR, and G-sec yields.
- ✓ **USD/INR, EUR/INR, GBP/INR, and JPY/INR** are traded on **NSE** and **MCX-SX**.
- ✓ Interest rate futures are legal but not yet widely adopted in India.



# UNIT 25

## Treasury and Asset-Liability Management

### STRUCTURE

- 25.0 Objectives
- 25.1 Meaning of Asset-Liability Management (ALM)
- 25.2 Liquidity Risk and Interest Rate Risk
- 25.3 Role of Treasury in ALM
- 25.4 Use of Derivatives in ALM
- 25.5 Credit Risk and Credit Derivatives
- 25.6 Transfer Pricing
- 25.7 Policy Environment

*Let Us Sum Up*



# Asset-Liability Management

The risks arise out of mismatch of assets and liabilities of the bank and asset-liability management is managing such balance sheet risks.

The risks, if not controlled, may result in negative spreads or in erosion of net worth. ALM is therefore defined as protection of net worth of the bank.

# Liquidity Risk

As we have seen earlier, liquidity and interest rate are two sides of the same coin, as the liquidity risk translates into interest rate risk, when the bank has to recycle the deposit funds or rollover a credit on market determined terms.

However, banks are extra sensitive to liquidity risks, as they cannot afford to default or delay meeting their obligations to depositors and other lenders.

Even suspicion of pressure over a bank's liquidity may prompt a run on the bank, or indeed, threaten the very survival of the bank.

Hence special attention is paid to liquidity, in particular short-term liquidity (intra-day to one month) to ensure funds are promptly made available when they are needed.

RBI has prescribed time bands (Next day, 2 to 7 days, 8 to 14 days, 15 to 30 days, etc.) for measuring and monitoring liquidity gaps. ALM process involves plotting of assets and liabilities maturity wise in time buckets and measuring the gap between assets and liabilities maturing in a specific time period. Liquidity risk is reflected as maturity mismatch – which is the gap in cash inflow and outflow. The risk is not being able to find enough cash, or cash at acceptable rate of interest, to fund the gap.

RBI from time to time issues detailed guidelines for managing ALM risks. RBI is more particular about short-term liquidity, ranging from intra-day to one month. Current guidelines stipulate that the net cumulative negative mismatches during the next day, 2-7 days, 8-14 days and 15-28 days buckets should not exceed 5%, 10%, 15% and 20% of the cumulative cash outflows in the respective time buckets. Banks are required to provide in their Liquidity Management Policy, contingency measures to meet any shortfall in liquidity. The contingency measures may include stand-by credit lines from other

## Interest Rate Risk

Interest rate risk arises when interest earnings are not adequate to set off interest payments due in a given period, even if the book value of the asset equals that of the liability, owing to a change in market rates of interest.

Net Interest Income (NII) of the bank is the difference between interest earnings and interest payments in a given accounting period.

Hence interest rate risk may be defined as the risk of erosion of NII account of interest rate movements in the market.

In a hypothetical situation, let us assume that the Bank has mobilized deposits of Rs. 100 cr., with average maturity of 6 months, at 5% interest. Let us also assume that the bank invested the amount in a fixed interest loan payable after 5 years at 7% p a. the NII is a clear 2% or Rs. 2 cr. per year.

The deposits mature after 6 months and need to be replaced or recycled at current market rate, say, at 6% as interest rates have risen by that time. The interest on loan continues to be 7%, hence NII for second half of the year is reduced by 1%. If we assume that the deposits become even costlier after next 6 months, demanding renewal at market rate of say, 8%, the NII actually becomes negative by 1%. However, if deposit rates fall by 2%, the NII correspondingly rises for the specific period.

RBI stipulates capital adequacy requirement for market risk, which includes interest rate mismatches. Capital is also to be provided for any derivatives (forwards, options and swaps) used to bridge such mismatches. RBI is recommending simplified approach under Basel 3, for determining the capital requirement for derivative instruments for Banks which handle a range of sophisticated derivative products like options, options with zero cost structures etc. Intermediate approaches like Delta-Plus Method and Scenario Approach have been recommended for Banks which write options and deal in other sophisticated derivative products.

## Experience of ALM in Indian Scenario

Depositors are always comfortable with fixed rate of interest. Bank like SBI and IDBI in the past introduced deposit schemes linked to floating rate interest, but it had not found the flavour of the depositors. Hence, these products were withdrawn. Under the circumstances, in case, the deposit rate goes up subsequent to placement of deposit by the depositors, the depositor would come for premature extension of the deposit and get the enhanced interest rate. Banks also do not charge any penalty for such extension.

## Experience of ALM in Indian Scenario

But in the advances side, RBI has introduced the MCLR system from 1s April, 2016, which is a floating rate of interest. Hence, under the interest rate falling scenario, the banks' NIM would be come down since the deposits are at fixed rate and advances are at floating rate.

In case, interest rate goes up, the banks may not actually get the full benefit of interest rate hike, as most of the depositors would come premature extension of the deposit. By this, the interest rate going up is nullified and the risk is passed on to the banks.

## Relationship Between Treasury And ALM As Under:

As the balance sheet of a bank carries enormous market risk (in addition to credit risk), but the banking operation itself is confined to accepting deposits, and extending credit to needy borrowers, besides miscellaneous payment services. It is Treasury which operates in financial markets directly, establishing a link between core banking functions and market operations. Hence the market risk is identified and monitored through Treasury.

Treasury, while taking trading positions in forex and securities markets, is also exposed to market risk on its own creation. Sometimes the risks are compensatory in nature and help bridge the mismatches on banking side. The Treasury may therefore hedge only residual risk.

## Relationship Between Treasury And ALM As Under:

As the markets develop, many credit products are being substituted by treasury products. For instance, bank may subscribe to commercial purpose, instead of extending working capital to an entity. Treasury are marketable and hence liquidity can be infused in times of need. Treasury also monitors exchange rate and interest rate movements in the markets, and hence it is much easier to administer such risks through treasury operations.

## Use of Derivatives In ALM

- ✓ Derivative instruments are useful in managing the liquidity and interest rate risks, as also in structuring new products which help overcome market risk to a large extent.
- ✓ Derivatives replicate market movements, and hence can be used to counter the risks inherent in regular transactions.
- ✓ For instance, if we are buying a stock which is highly sensitive to market movements, we can sell index futures as an insurance/hedge against fall in stock prices. The advantage in derivatives is that the requirement of capital is very small, and largely there is no deployment of funds (except in case of exchange traded instruments, where there is a margin requirement - but it is only a fraction of notional amount).
- ✓ Derivatives can be used to hedge high value individual transactions, or aggregate risks as reflected the asset-liability mismatches. In the latter case, a dynamic management of hedge is necessary as the composition of assets and liabilities is always changing.

## Credit Risk and Credit Derivatives

As we have seen, Treasury is mostly concerned with market risk. Credit risk in treasury business is only with respect to counterparty dealings, contained by exposure limits and risk management norms. In normal course, treasury operations are untouched by the credit risk present in bank's lending business.

## Credit Derivatives

Credit derivatives (CD) help the issuer diversify the credit risk and use the capital more efficiently. The CD is a transferable instrument, though the market for CDs is not very liquid. The CD products are still emerging and various covenants related to the transaction are incorporated in the ISDA Master Agreement for Credit Derivatives.

***The eligible entities under market-makers and users categories are as under:***

- **Market Makers**
- **Users**



## **Market makers:**

Commercial Banks, standalone Primary Dealers (PDs), Non- Banking Financial Companies (NBFCs) having sound financials and good track record in providing credit facilities and any other institution specifically permitted by the Reserve Bank. Insurance companies and Mutual Funds would be permitted if permitted by their regulators.

## **Users:**

Commercial Banks, PDs, NBFCs, Mutual Funds, Insurance Companies, Housing Finance Companies, Provident Funds, Listed Corporates, All India Financial Institutions namely, Export Import Bank of India (EXIM), National Bank for Agriculture and Rural Development (NABARD), National Housing Bank (NHB) and Small Industries Development Bank of India (SIDBI), Foreign Institutional Investors (FIIS) and any other institution specifically permitted by the Reserve Bank.

## Credit Derivatives

RBI has prescribed eligibility norms for market makers and has also prescribed detailed operational guidelines and capital adequacy norms for Credit Default Swaps.

## Transfer Pricing

Transfer pricing refers to fixing the cost of resources and return on assets of the bank in a rational manner.

The treasury notionally buys and sells the deposits and loans of the bank, and the price at which the treasury buys and sells forms the basis for assessing profitability of banking activity.

The treasury determines the buy/sell prices on the basis of market rates of interest, the cost of hedging market risk and the cost of maintaining reserve assets of the bank.



## Policy Environment

**The asset-liability management will be effective, only if there is a strong policy foundation.** Though in general we describe it as ALM Policy, the policy should aim at aggregate risk of the bank and should achieve coordination between different departments of the bank and treasury. **An Integrated Risk Management Policy, bearing upon Market Risk of the Bank, should ideally have the following components:**

- **ALM Policy** prescribes the composition of Asset Liability Management Committee (ALCO) and operational aspects of ALM, including risk measures, monitoring of risks, risk neutralization, product pricing, management information systems and more importantly how to improve and/or maintain the existing NIM, etc.
- **Liquidity Policy** prescribes minimum liquidity to be maintained, funding of reserve assets, limits on exposure to money market, contingent funding, inter- bank committed credit lines etc.

## Policy Environment

- **Derivatives Policy** prescribes norms for use of derivatives, capital allocation, restrictions on derivative trading, valuation norms, exposure limits etc.
- **Investment Policy** prescribes the permissible investments, norms re-credit rating and listing, SLR and Non-SLR investments, private placement, trading in securities and repos, classification and valuation of investments, accounting policy etc.
- **Composite Risk Policy for Foreign Exchange and Treasury** prescribes norms for merchant and trading positions, securities trading, exposure limits, limits on intra-day and overnight positions, stop-loss limits, periodical valuation of trading positions etc.
- **Transfer Pricing Policy** prescribes the methodology, spreads to be retained by treasury, segregation of administrative costs and hedging costs, allocation of costs to branches/other departments of the bank





**JAIIB CAIIB**  
PREP

SUBSCRIBE NOW



# JAIIB CAIIB P

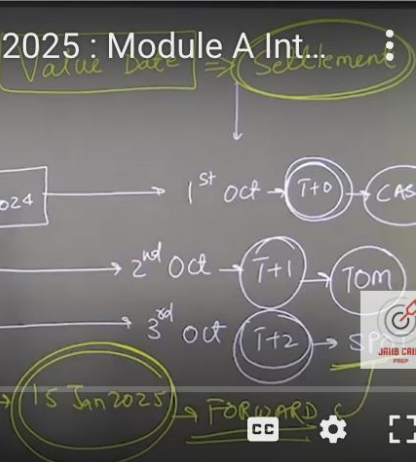
@JAIIB\_CAIIB\_Prep\_By\_MaheshSir ·

Welcome to JAIIB CAIIB Prep – You can find all the study materials, notes, and videos on [edushield.page.link/SjVn](https://edushield.page.link/SjVn) and 2 more

Customize channel Manage

Home Videos Shorts Live Playlists Pos

CAIIB BFM Marathon 2025 : Module A Int...



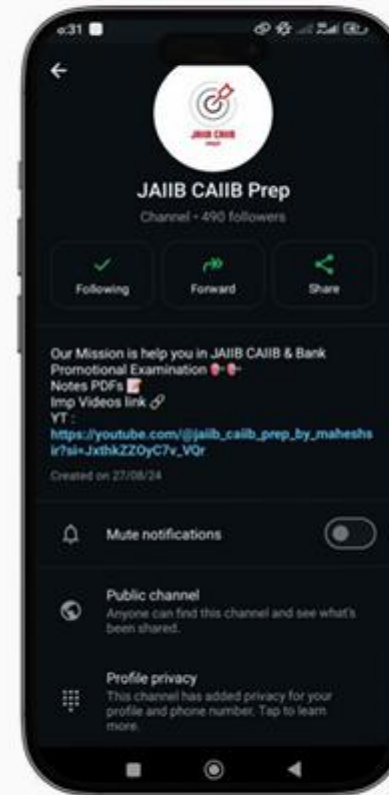
27:08 / 11:40:31



# Download our app



<https://www.jaiibcaiibprep.com>



Follow to our  
**WhatsApp**  
Channel

*Subscribe to our Youtube Channel **JAIIB CAIIB PREP** now !*

# THANK YOU!



@JAIBCAIIBPREPOFFICIAL